

```
In [39]: import nltk
import math
from nltk.corpus import stopwords
from nltk.tokenize import word_tokenize
nltk.download('punkt')
nltk.download('stopwords')
```

```
[nltk_data] Downloading package punkt to
[nltk_data]   C:\Users\mestr\AppData\Roaming\nltk_data...
[nltk_data]   Package punkt is already up-to-date!
[nltk_data] Downloading package stopwords to
[nltk_data]   C:\Users\mestr\AppData\Roaming\nltk_data...
[nltk_data]   Package stopwords is already up-to-date!
```

Out[39]: True

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In [40]: documentA = 'Jupiter is the largest Planet'
documentB = 'Mars is the fourth planet from the Sun'
```

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In [41]: documents = [documentA, documentB]
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In [42]: stop_words = set(stopwords.words('english'))
```

```
In [44]: processed_docs = []

for doc in documents:
    doc = doc.lower()
    words = word_tokenize(doc)
    filtered_words = [w for w in words if w not in stop_words]
    processed_docs.append(filtered_words)

print("Processed Documents:", processed_docs)
```

```
Processed Documents: [['jupiter', 'largest', 'planet'], ['mars', 'fourth', 'plane
t', 'sun']]
```

```
In [51]: vocab = set()
for doc in processed_docs:
    vocab.update(doc)

vocab = list(vocab)
print(vocab)
```

```
['mars', 'sun', 'largest', 'planet', 'fourth', 'jupiter']
```

```
In [59]: def compute_tf(doc):
    tf_dict = {}
    doc_len = len(doc)
    for word in vocab:
        tf_dict[word] = doc.count(word) / doc_len
    return tf_dict

tf_docs = [compute_tf(doc) for doc in processed_docs]
print(tf)
```

```
{'mars': 0.25, 'sun': 0.25, 'largest': 0.0, 'planet': 0.25, 'fourth': 0.25, 'jupi
ter': 0.0}
```

```
In [61]: def compute_idf(docs):
    N = len(docs)
    idf_dict = {}

    for word in vocab:
        count = sum(1 for doc in docs if word in doc)
        idf_dict[word] = math.log(N / (count + 1)) + 1 # smoothing

    return idf_dict

idf = compute_idf(processed_docs)
print(idf)

{'mars': 1.0, 'sun': 1.0, 'largest': 1.0, 'planet': 0.5945348918918356, 'fourth': 1.0, 'jupiter': 1.0}
```

```
In [62]: def compute_tfidf(tf, idf):
    tfidf = {}
    for word in vocab:
        tfidf[word] = tf[word] * idf[word]
    return tfidf

tfidf_docs = [compute_tfidf(tf, idf) for tf in tf_docs]
```

```
In [64]: print("\nVocabulary:", vocab)

Vocabulary: ['mars', 'sun', 'largest', 'planet', 'fourth', 'jupiter']
```

```
In [65]: print("\nTF for each document:")
    for i, tf in enumerate(tf_docs):
        print(f"Document {i+1}:", tf)

TF for each document:
Document 1: {'mars': 0.0, 'sun': 0.0, 'largest': 0.3333333333333333, 'planet': 0.3333333333333333, 'fourth': 0.0, 'jupiter': 0.3333333333333333}
Document 2: {'mars': 0.25, 'sun': 0.25, 'largest': 0.0, 'planet': 0.25, 'fourth': 0.25, 'jupiter': 0.0}
```

```
In [66]: print("\nIDF values:")
    print(idf)

IDF values:
{'mars': 1.0, 'sun': 1.0, 'largest': 1.0, 'planet': 0.5945348918918356, 'fourth': 1.0, 'jupiter': 1.0}
```

```
In [67]: print("\nTF-IDF for each document:")
    for i, tfidf in enumerate(tfidf_docs):
        print(f"Document {i+1}:", tfidf)

TF-IDF for each document:
Document 1: {'mars': 0.0, 'sun': 0.0, 'largest': 0.3333333333333333, 'planet': 0.19817829729727854, 'fourth': 0.0, 'jupiter': 0.3333333333333333}
Document 2: {'mars': 0.25, 'sun': 0.25, 'largest': 0.0, 'planet': 0.1486337229729589, 'fourth': 0.25, 'jupiter': 0.0}
```

```
In [ ]:
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